MOHAMED CHERFA
Cybersecurity Engineer | Reverse Engineering & Penetration Testing

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PROFILE

An engineer, specializing in offensive cybersecurity (reverse engineering, penetration testing). Ranked in the top 0.7% on Root-Me and holder of the Hacker badge on Hack The Box .

EDUCATION

COMPUTER ENGINEERING DEGREE | National Higher School of Computer Science (ESI) | Algiers | 2021 - 2026
Specialization: Information Systems and Technologies - Cybersecurity Track
BACCALAUREATE IN MATHEMATICS | 2021 | High Honors (17.91/20)

PROFESSIONAL EXPERIENCE

DEVELOPER | Plast Prestige | Blida | Aug 2026–Present

- Developed a secure, custom ERP system from scratch.
- **Technologies:** React/TypeScript, Node.js/Express, PostgreSQL.

Security Auditor & HR Software Developer | Le Printemps Algeria | 2026

-Performed a security audit of an existing HR application, including vulnerability analysis, penetration testing, and the delivery of a detailed recommendations report.
-Developed a new secure HR system from scratch.
-Technologies used: React/TypeScript, Node.js/Express, MongoDB, Electron, Socket.io.

WEB DEVELOPMENT INSTRUCTOR | Progress Academy, Blida | 2025

- Trained 20+ students in modern web technologies (React, Node.js, JavaScript) and security best practices
- Created educational materials and mentored hands-on practical projects

REVERSE ENGINEER FREELANCE | Fiverr

- BSON file deobfuscation for sensitive data recovery
- Tools: Ghidra, Binwalk, Python, JADX, APKTool

IT INFRASTRUCTURE INTERN | Mobilis (Telecom Operator), Algiers | Sept - Oct 2024

- Virtualized 6 servers using VMware vSphere ESXi (web, database, messaging services)
- Automated monitoring with custom Python scripts and Nagios integration

- Technical documentation and system team training

CO-MANAGER DEVELOPMENT DEPARTMENT | Code&Share Club, ESI | 2022 - 2023

- Mentored 15+ members on web projects and organized 8 technical workshops
- Coordinated 3 major events (Internal Hackathon, Code Sprint, mentoring sessions)

TECHNICAL PROJECTS

FINAL YEAR PROJECT - Multi-Modal Behavioral Biometric System | LMCS Lab, ESI | 2025-2026
Continuous authentication system based on behavioral analysis (keystroke dynamics, mouse movements, interaction patterns)
- technologies: Python, Scikit-learn/TensorFlow, React, Flask | Supervisor: Dr. S. MILI

PetroMonitore | Academic Project , Sonatrach Partnership Project | 2024-2025
Web application for oil drilling operations monitoring for Sonatrach (Algeria's national oil company) (React/Django)
- Lead Frontend & UI/UX: real-time dashboard, alerts, automated reports
- Team of 4 | Technologies: React, Material-UI, Django REST, PostgreSQL

BANKI | Academic Project ESI | 2022-2023
Smart comparison platform for Algerian banking offers (React/Express)
- Lead Frontend & UI/UX: responsive interface, rating system, personalized recommendations
- Team of 7 | Technologies: React, Express.js, MongoDB

CTFs & COMPETITIONS

Root-Me | Top 0.7% | Insider Badge
- 180+ challenges solved (Reverse Engineering, Cryptography, Web, Forensics, Pwn)
- Expertise: 42% of hard reverse engineering challenges solved | Profile: root-me.org/mo7wf

Hack The Box | Hacker Badge | 20+ machines pwned, 30+ challenges solved
- Mastery: reconnaissance, exploitation, privilege escalation, post-exploitation

TECHNICAL SKILLS

Reverse Engineering: Ghidra, IDA Pro, x64dbg, dnSpy, Frida, Radare2, JADX, Binwalk
Pentest & Security: Burp Suite, Metasploit, Nmap, Wireshark, SQLmap, Hashcat, John the Ripper
Development: Python (advanced), C/C++, Java, JavaScript/React, Bash, SQL
Systems & Networks: Linux (Kali, Ubuntu, CentOS), VMware, Docker, TCP/IP, Cisco

LANGUAGES

Arabic (Native) | French (Fluent - C1, TCF) | English (Fluent - C1)